Understanding What Drives Spotify Premium Willingness

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Problem & Hypothesis

Problem:

Which user behaviors most strongly predict willingness to subscribe to Spotify Premium?

Understanding specific user behaviors and demographic factors that predict this conversion is one of the most important things for a B2C company like Spotify.

Hypothesis:

High versatility of listening contexts and satisfaction with the recommendation system will strongly predict Premium conversion.

Data & Preparation

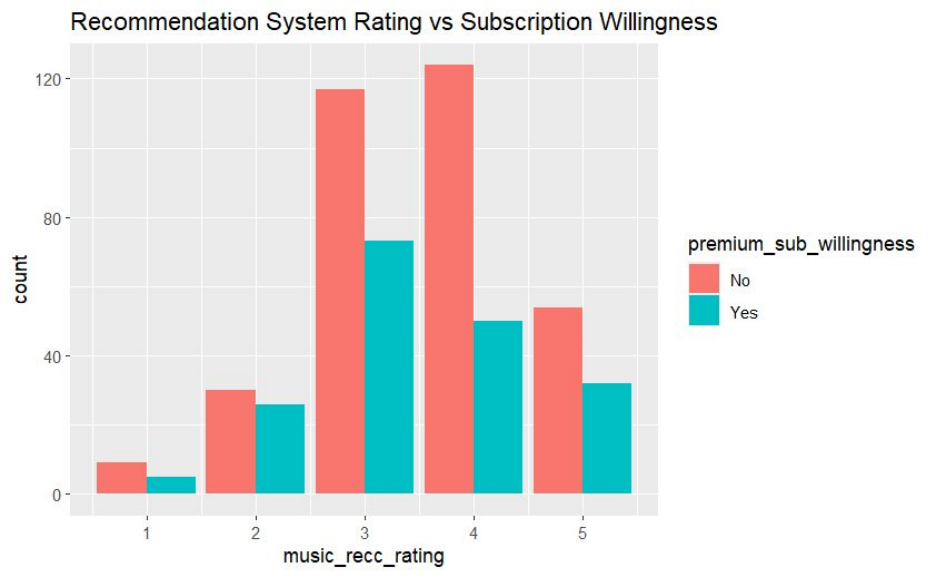
Data:

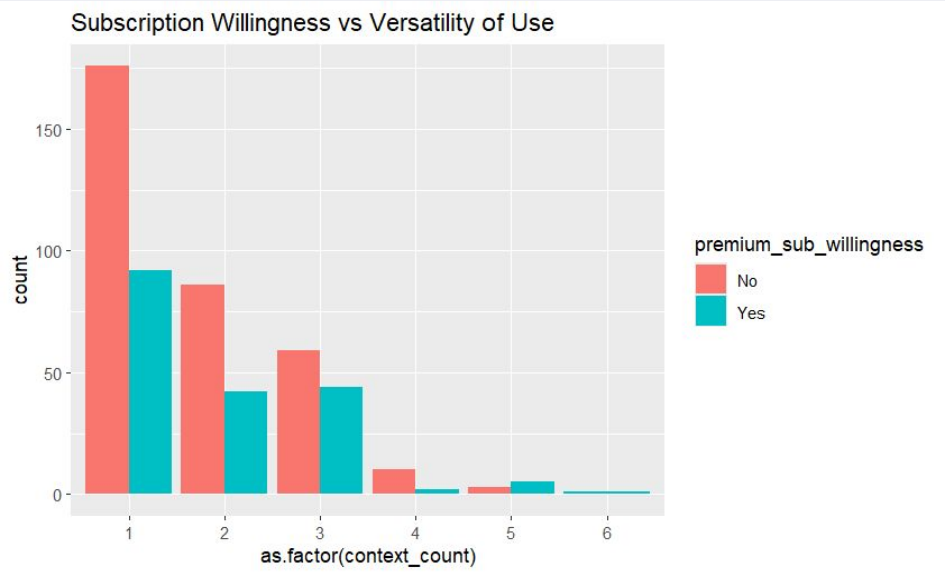
- Source: Kaggle "Spotify User Behavior Dataset" (520 observations)
- Captures a balanced mix of free/premium users and deeply specific behavioral metrics
- Target variable: premium_sub_willingness (Yes/No)
- 4 features selected: Age, music_recc_rating, context_count, spotify_usage_period

Preparation:

- Basic data preprocessing like cleaning and checking for missing values
- Reordered ordinal variables chronologically
- Cleaned up messy, comma-separated string data regarding listening habits
- Made histograms to visualize distributions of each feature

Preliminary & Exploratory Data Analysis





No clear difference in proportion of users willing to subscribe with higher ratings or higher versatility of use.

- For recommendation ratings, the proportion of Yes didn't increase as ratings got higher.
- For listening versatility, users with more contexts weren't any more likely to subscribe either.

Hypothesis Testing Results

Feature	Test	p-value	Significant?
Recommendation Rating	t-test	0.1496	No
Context Versatility	t-test	0.1592	No
Age	Chi-square	0.1184	No
Usage Period	Chi-squre	0.000228	Yes

- Three of the four features showed no significant relationship with subscription willingness. Only usage period was significant with a p-value of 0.000228.
- Only longevity/usage period matters. My hypothesis was wrong.

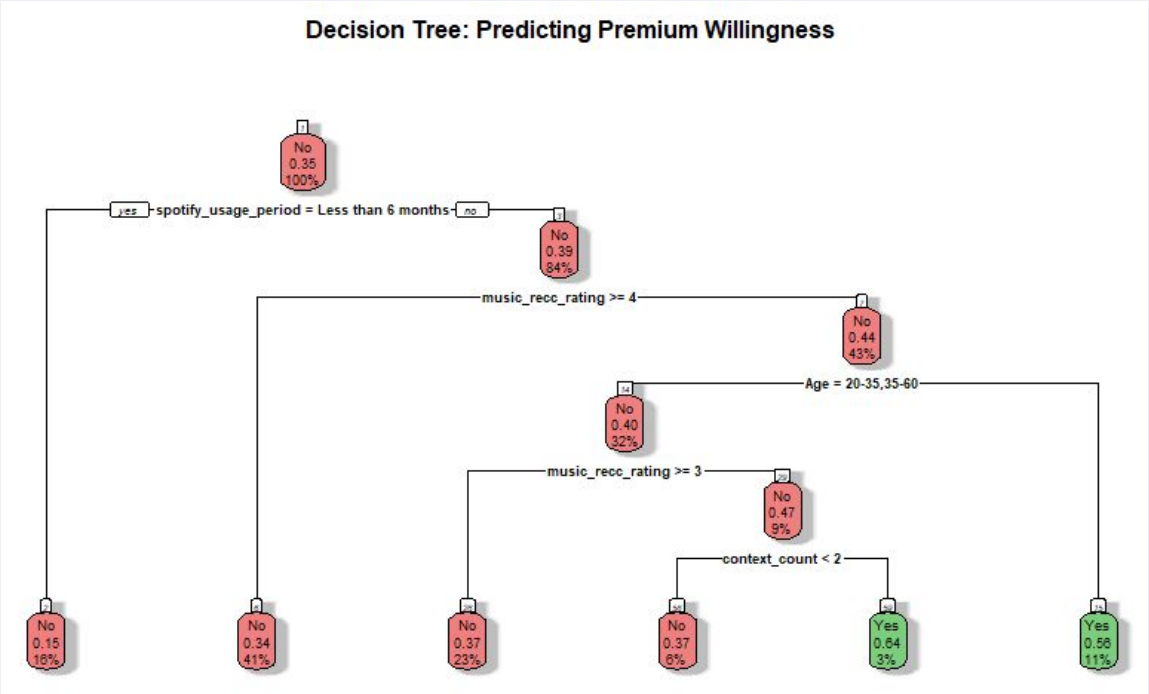
Decision Tree

First split: spotify_usage_period

- Confirms hypothesis testing

Accuracy: 64.42%

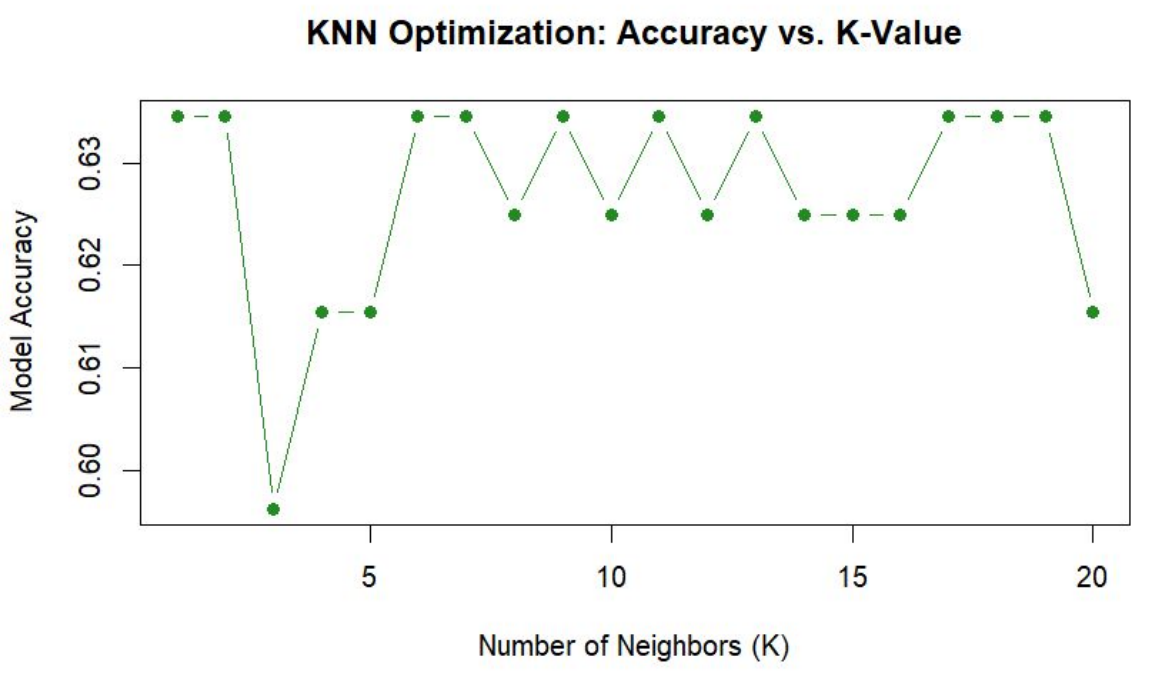
Great at predicting "No" (60 True Negatives), poor at finding subscribers (only 7 True Positives, 32 False Negatives)



	Actual: No	Actual: Yes
Predicted: No	60	32
Predicted: Yes	5	7

KNN

- Accuracy: 65.38%
- Captured 13 True Positives, 26 False Negatives
- Same core limitation as Decision Tree: feature set is too weak to reliably identify subscribers



	Actual: No	Actual: Yes
Predicted: No	55	26
Predicted: Yes	10	13

Conclusion

Results

- Hypothesis disproved: satisfaction and versatility don't drive conversion, loyalty does.

Spotify

- Focus retention strategies on keeping free users engaged long-term.

Next Steps

- Get a better dataset (listening hours, trial history, ad skip rates)
- Try Random Forest for better feature importance

Thank You